Supplementary Material for:
Regional Inflation Spillovers and Monetary Policy Design

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This supplementary material provides the data link for the constructed regional CPI index covering nine economic regions, along with a comprehensive set of robustness exercises supporting the paper's main findings. It includes alternative model specifications, additional estimations in both the frequency and time domains, city-level analyses, and a wide range of sensitivity checks across forecast horizons, lag structures, and spatial aggregation schemes.

A. Data Basis

This study employs monthly Consumer Price Index (CPI) data with base December 2021=100 for Peru's 25 largest cities, published by the National Institute of Statistics and Informatics (INEI), spanning January 2002 to December 2024. To ensure temporal consistency, we retroactively extended the December 2021 base backward by applying monthly percentage changes from the previous 2009-based index, creating a unified 23-year time series that captures both structural changes and cyclical variations in regional price dynamics. Based on these 25 city-level CPI series, we construct the regional price indices for the nine economic regions used in the paper. This dataset is available at https://github.com/jaguimu/Inflation-Spillovers-and-Monetary-Policy-Design

B. Regional Inflations

This study employs monthly Consumer Price Index (CPI) data with base December 2021=100 for Peru's 25 largest cities, published by the National Institute of Statistics and Informatics (INEI), spanning January 2002 to December 2024. To ensure temporal consistency, we retroactively extended the December 2021 base backward by applying monthly percentage changes from the previous 2009-based index, creating a unified 23-year time series that captures both structural changes and cyclical variations in regional price dynamics.

Following the established framework of Winkelried and Gutierrez (2015), we adopt the economic classification of Peruvian cities into nine economic regions proposed by Gonzales de Olarte (2003). This classification, summarized in Table 1, is grounded in both historical considerations-as regions comprise contiguous departments with shared cultural and administrative legacies-and economic factors, including market articulation, trade integration, and production specialization. The nine-region framework effectively captures Peru's economic geography, from Lima's urban concentration to the Amazon's resource extraction, the coast's agricultural production, and the highlands' mining activities.

Table 1: Economics Regions in Peru

Region	Cities (weights)
1	Lima (66.02)
2	Piura (2.01), Tumbes (0.56)
3	Chiclayo (2.55), Cajamarca (1.14), Chachapoyas (0.14)
4	Trujillo (4.60), Chimbote (2.05), Huaraz (0.58)
5	Ica (1.31), Ayacucho (0.70), Huancavelica (0.19)
6	Arequipa (4.94), Moquegua (0.30), Tacna (1.55), Puno (0.76)
7	Huánuco (0.81), Cerro de Pasco (0.34), Huancayo (1.91)
8	Abancay (0.34), Cusco (2.72), Puerto Maldonado (0.48)
9	Iquitos (1.99), Moyobamba (0.64), Pucallpa (1.30)

Notes: Classification of Peruvian Departments into 9 economic regions, following Gonzáles de Olarte (2003). Numbers in parentheses indicate the weights used for national inflation calculations by INEI. The weights reflect the national expenditure shares utilized in computing inflation using the 2009 base year.

To construct regional price indices, we employ a weighted additive approach where each city's CPI is weighted according to its respective share in national consumption patterns[1]. The aggregated regional CPI is calculated as:

$$CPI_{r,t} = \frac{\sum_{i \in r} w_i . CPI_{i,t}}{\sum_{i \in r} w_i} \quad (1)$$

where $CPI_{r,t}$ denotes the Consumer Price Index for region r at time t, w_i is the weight of department i within región r, and $CPI_{i,t}$ is the Consumer Price Index of department i at time t. This weighting scheme ensures that regional inflation measures accurately reflect the relative economic importance of each city within its region while maintaining consistency with national inflation calculations.

Based on the regional Consumer Price Indices, monthly inflation for region r at time t is computed as:

$$x_{r,t} = 100 \left(\log CPI_{r,t} - \log CPI_{r,t-1} \right) \quad (2)$$

This log-difference specification provides several analytical advantages: it yields approximately percentage changes for small variations, ensures stationarity properties essential for spillover analysis, and facilitates direct comparison across regions with different price levels.

Figure 1 illustrates monthly inflation across Peru's nine economic regions, revealing both striking synchronization and notable heterogeneity in regional price dynamics. Despite considerable differences in volatility and amplitude across regions, the time series exhibit a remarkably synchronized pattern, indicating the influence of shared macroeconomic and structural factors on regional price formation. This co-movement provides preliminary evidence of substantial spillover effects that our formal analysis will quantify.

Three major inflationary episodes, highlighted in Figure 1, demonstrate how external shocks propagate through Peru's regional system:

[1] Although INEI updated the CPI base to December 2021=100 and now publishes indices using this new base, the revised expenditure weights by department are not publicly available. As a result, regional aggregation continues to rely on the weights from the 2009 base.

- The 2007-2009 Global Commodity Boom: The first major episode coincided with the global commodity price surge, which drove up food and energy costs worldwide. In Peru, this translated to synchronized inflation increases across all regions, with food-producing areas (Regions 3 and 7) experiencing particularly pronounced spikes that subsequently transmitted to other regions. This episode illustrates how global supply shocks can amplify regional disparities before eventually synchronizing national inflation dynamics, as emphasized in broader EME contexts by Moreno (2010).

- The 2017 *Coastal El Niño* Event: The sharp but short-lived spike in March 2017 aligns with the Coastal El Niño phenomenon, which severely disrupted transportation and logistics networks in northern and central Peru. This event provides a natural experiment in regional shock transmission: the initially localized supply disruptions in coastal regions rapidly propagated inland through supply chain linkages, leading to temporary but significant price pressures across the national economy (Yglesias-Gonzáles et al., 2023). The rapid diffusion and subsequent normalization demonstrate both the vulnerability and resilience of Peru's regional price system.

- The 2021-2023 Pandemic and Geopolitical Inflation: The most recent episode reflects the complex interaction of pandemic-related supply chain disruptions and elevated international food and energy costs stemming from the Russia-Ukraine conflict. Unlike previous episodes that originated domestically or affected specific sectors, this global shock simultaneously impacted all regions through multiple transmission channels: supply chain fragmentation, energy price pass-through, and exchange rate depreciation (Central Reserve Bank of Peru, 2023). The synchronized response across regions, despite varying degrees of international integration, underscores the increasing interconnectedness of Peru's regional economies.

The observed temporal co-movement of regional inflation—particularly pronounced during major inflationary episodes—provides compelling preliminary evidence of meaningful domestic spillovers. Simple correlation analysis, shown in Table 2, reveals that all regional inflation series are positively correlated, with an average correlation of 0.59 across all regional pairs. These correlations, while substantial, vary meaningfully across region pairs, suggesting that spillover effects are neither uniform nor complete, motivating the need for directional spillover analysis.

Table 2: Correlation Matrix

	Lima	Region 2	Region 3	Region 4	Region 5	Region 6	Region 7	Region 8	Region 9
Lima	1.000	0.699***	0.750***	0.754***	0.734***	0.578***	0.713***	0.376***	0.584***
Region 2		1.000	0.764***	0.777***	0.613***	0.535***	0.605***	0.319***	0.595***
Region 3			1.000	0.762***	0.673***	0.554***	0.690***	0.411***	0.618***
Region 4				1.000	0.709***	0.579***	0.709***	0.425***	0.579***
Region 5					1.000	0.596***	0.713***	0.499***	0.612***
Region 6						1.000	0.602***	0.662***	0.441***
Region 7							1.000	0.569***	0.559***
Region 8								1.000	0.369***
Region 9									1.000

Notes: This table reports the correlation coefficients of regional inflation rates along with their statistical significance: *** denotes significance at the 1% level, ** at the 5% level, and * at the 10% level.

Summary statistics reported in Table 2 reveal important heterogeneity in regional inflation characteristics. Lima (Region 1) exhibits the lowest level of inflation volatility (standard deviation of 0.33), despite its central role, consistent with its diversified economic base and sophisticated distribution networks. In contrast, regions such as Region 2 (0.52), Region 4 (0.48), and Region 8 (0.48) show higher volatility, likely reflecting greater exposure to agricultural supply disruptions, seasonal demand fluctuations, and localized logistical constraints. In terms of distributional shape, Lima also displays the lowest excess kurtosis (0.80), suggesting relatively stable inflation dynamics. By comparison, most other regions exhibit more pronounced kurtosis, especially Regions 2 and 3 (2.78 and 2.30, respectively), both agricultural producers. These values indicate a higher frequency of extreme inflation realizations, consistent with the presence of price spikes in food-related sectors.

The presence of occasional extreme values-particularly in agricultural and resource-dependent regions-motivates our choice of spillover methodologies that can accommodate non-normal distributions and capture directional transmission effects. The combination of strong co-movement with persistent regional heterogeneity suggests a complex spillover structure where some regions consistently transmit shocks while others primarily receive them-precisely the type of asymmetric relationship that modern spillover indices are designed to quantify.

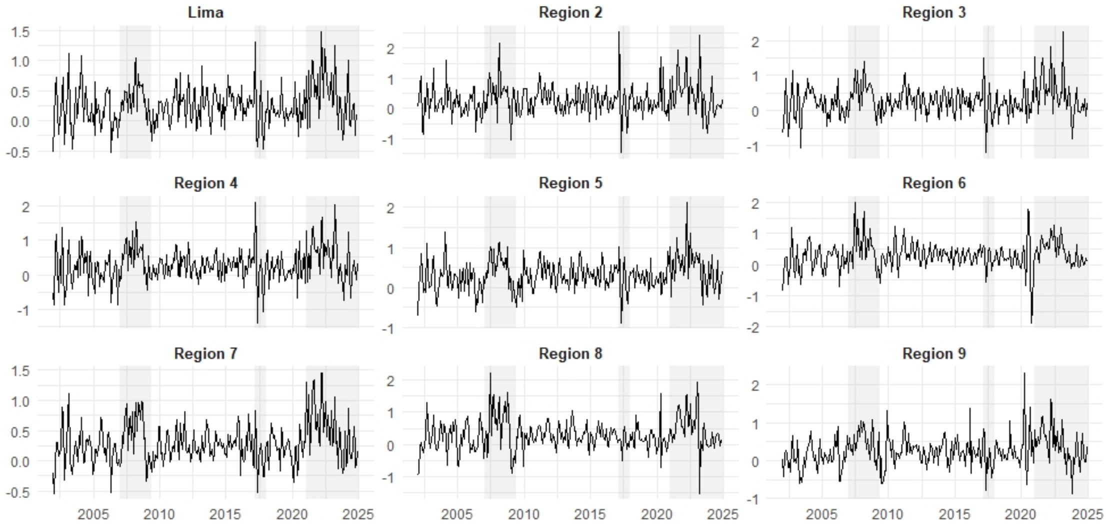
Finally, Table 3 also presents the results of the Augmented Dickey–Fuller (ADF) and Phillips–Perron (PP) unit root tests applied to the monthly regional inflation series. In all cases, the null hypothesis of nonstationarity is rejected. These results justify the use of a generalized VAR framework to compute spillover indices following DY and BK approaches.

Table 3: Summary statistics

	Lima	Region 2	Region 3	Region 4	Region 5	Region 6	Region 7	Region 8	Region 9
Mean	0,254	0,296	0,28	0,282	0,318	0,289	0,286	0,296	0,254
Std. Dev.	0,331	0,281	0,436	0,404	0,388	0,449	0,338	0,478	0,414
Skewness	0,88	0,731	0,801	0,482	0,357	0,609	0,738	0,706	0,745
Kurtosis	3,28	2,208	2,599	1,534	1,559	2,407	2,774	2,805	2,606
ADF	-3.655***	-3.224***	-3.599***	-3.995***	-4.146***	-4.922***	-3.978***	-3.655***	-3.456***
PP	-13.017***	-12.493***	-12.005***	-13.677***	-12.316***	-10.014***	-11.044***	-10.644***	-11.367***

Notes: The table reports the test statistics from the Augmented Dickey-Fuller (ADF) and Phillips-Perron (PP) unit root tests. For the ADF test, an intercept was included as the deterministic component, and the lag length was selected based on the t-statistic criterion. *** denotes significance at the 1% level, ** at the 5% level, and * at the 10% level.

Figure 1: Monthly regional inflations



C. Robustness Analysis

To assess the reliability and generalizability of our findings, we conduct an extensive robustness analysis that evaluates the sensitivity of the spillover estimates across alternative model specifications, temporal aggregations, spatial resolutions, and methodological choices. This multi-dimensional approach is essential given the policy relevance of our results for Peru's monetary framework and their potential applicability to other emerging economies with significant regional heterogeneity.

Our robustness analysis shows that the core spillover results remain highly stable across a broad set of alternative specifications and data treatments. Varying forecast horizons, lag lengths, and model structures leaves the regional spillover hierarchy largely unchanged, with Lima consistently acting as the main long-run transmitter and peripheral regions remaining net recipients. Additional checks using city-level data reproduce the same network structure, confirming that our findings are not driven by the chosen spatial aggregation. Likewise, alternative frequency decompositions—two-band and three-band—preserve the distinction between short-run supply-driven transmission and Lima's medium- and long-run influence linked to policy and structural dynamics. Taken together, the strong consistency of results across all dimensions reinforces the robustness, credibility, and broader applicability of our empirical conclusions.

C.1 Time-Domain Robustness Checks

Sensitivity to Model Specification Parameters

We begin by assessing the robustness of time-domain spillover estimates to key methodological choices that could potentially influence our conclusions. Following the approaches of Diebold and Yilmaz (2012), we systematically vary forecast horizons and lag lengths while monitoring the stability of regional spillover rankings and magnitudes.

Table 1 reports the sensitivity of spillover measures to forecast horizon variations, with horizons ranging from H=1 to H=12 months while maintaining a fixed lag length of L=1. Each cell presents the median value across horizons, with minimum and maximum values in brackets to illustrate the range of variation. The results demonstrate remarkable stability in regional spillover hierarchies: Lima consistently emerges as the dominant net transmitter across all forecast horizons, with median NET spillovers of 23.99 percentage points and a narrow range of 11.78 to 24.00 percentage points.

Similarly, Regions 8 and 9 maintain their roles as significant net recipients across all horizons, with relatively narrow ranges around their median values (-22.44 and -25.24 respectively). This consistency demonstrates that our key findings about regional transmission hierarchies are not artifacts of specific forecast horizon choices but reflect robust structural relationships in Peru's regional inflation network.

Table 1: Sensitivity of time domain spillovers to the forecast horizon

Region	TO	FROM	NET
Lima	100.5 [88.03, 100.51]	76.51 [76.25, 76.51]	23.99 [11.78, 24]
Region 2	75.72 [75.71, 80.5]	75.99 [74.75, 75.99]	-0.27 [-0.27, 5.75]
Region 3	78.75 [78.74, 81]	78.26 [74.95, 78.27]	0.48 [0.48, 6.05]
Region 4	83.72 [83.72, 87.56]	78.51 [76.15, 78.51]	5.21 [5.21, 11.4]
Region 5	81.16 [74.94, 81.16]	76.19 [73.13, 76.19]	4.97 [1.81, 4.97]
Region 6	72.62 [61.29, 72.63]	68.44 [67.92, 68.44]	4.18 [-6.62, 4.19]
Region 7	84.65 [81.5, 84.65]	75.53 [74.59, 75.54]	9.11 [6.91, 9.12]
Region 8	39.74 [32.96, 39.75]	62.19 [54.13, 62.19]	-22.44 [-22.86, -21.18]
Region 9	46.33 [46.32, 48.8]	71.57 [64.71, 71.57]	-25.24 [-25.25, -15.91]
TSI		73.69 [70.73, 73.69]	

Notes: This table shows the sensitivity of time domain spillovers to the forecast horizon (h = 1 to 12). Each cell reports the median value, with minimum and maximum across horizons in brackets. The lag of the VAR model is L=1, that was chosen by the BIC criterium.

Table 2 examines sensitivity to VAR lag length specifications, varying from L=1 to L=6 while holding the forecast horizon constant at H=3. The results confirm the robustness of our spillover rankings, with Lima's median NET spillover of 20.00 percentage points remaining substantially above all other regions across all lag specifications. The stability of these patterns suggests that our findings capture genuine economic relationships rather than statistical artifacts of particular model specifications.

Table 2: Sensitivity analysis of time domain spillovers with respect to VAR lag length

Region	TO	FROM	NET
Lima	96.67 [94.85, 100.43]	76.55 [75.82, 76.97]	20 [19.03, 23.94]
Region 2	76.6 [76.06, 78.03]	75.61 [74.62, 76.24]	1.19 [0.1, 2.99]
Region 3	79.72 [78.14, 80.54]	77.78 [76.65, 78.35]	2.01 [0.62, 3.18]
Region 4	85.56 [83.84, 87.12]	78.09 [76.72, 78.64]	7.85 [5.36, 9.47]
Region 5	80.93 [77.62, 82.34]	75.98 [75.08, 76.64]	4.81 [2.45, 5.7]
Region 6	71.63 [67.59, 72.13]	68.48 [67.18, 68.9]	2.8 [0.41, 3.76]
Region 7	84.5 [83.78, 84.69]	75.43 [74.83, 75.86]	8.83 [8.46, 9.87]
Region 8	39.98 [37.52, 41.55]	64.86 [61.93, 65.99]	-24.83 [-26.97, -22.62]
Region 9	47.53 [45.37, 49.18]	70.4 [68.78, 71.41]	-22.68 [-24.87, -21.2]
TSI		73.68 [72.89, 74.14]	

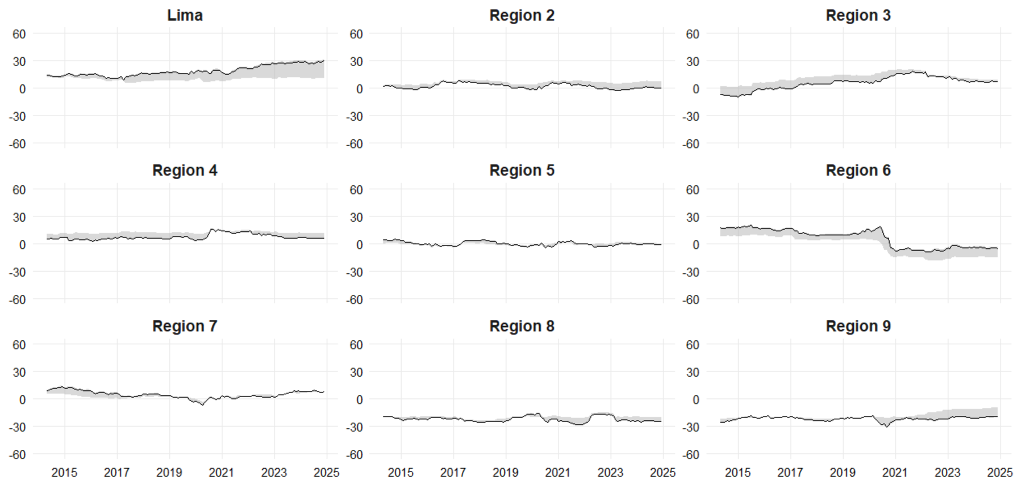
Notes: Each cell reports the median value of the time domain spillover measure across different lag lengths, with the minimum and maximum in brackets. Spillovers are computed using a forecast horizon of h = 3, varying the lag length of the VAR model from L = 1 to L = 6.

Dynamic Robustness Assessment

To assess the stability of our dynamic spillover estimates, we implement rolling-window analysis across forecast horizons H=1 to H=12. Figure 1 illustrates the evolution of NET spillovers, with the solid black line denoting the median trajectory across horizons and the shaded area capturing the full range between minimum and maximum estimates.

Lima's dynamic NET spillover evolution shows remarkable consistency, with the median closely tracking the upper bound throughout the sample period. This pattern reflects Lima's persistently strong transmission role across different forecast horizons. Notably, Lima's spillover intensity increases with horizon length, as evidenced by the lower bound (corresponding to H=1 and H=2) remaining substantially below the median. This horizon-dependent strengthening provides empirical support for our frequency-domain analysis, which explicitly demonstrates Lima's greater influence at longer cyclical periods.

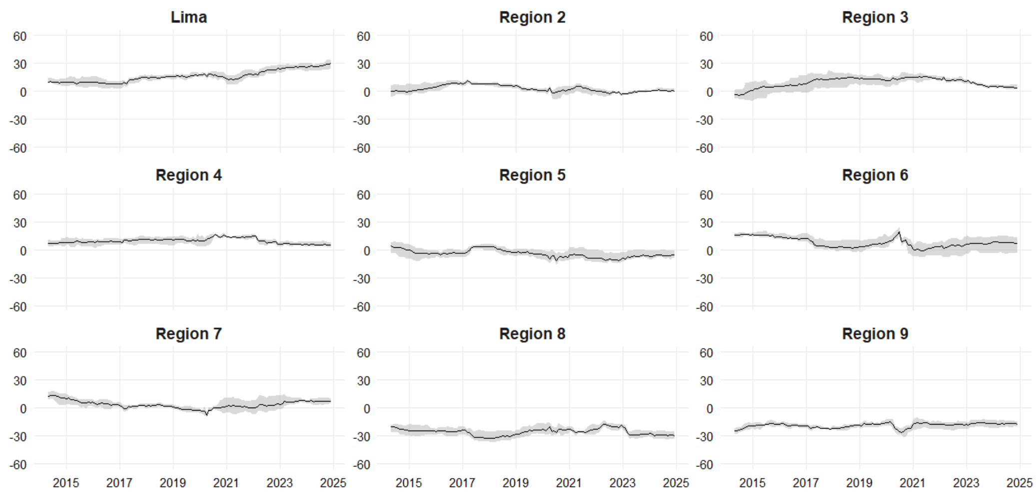
Figure 1: Robustness of Dynamic NET spillovers to Forecast Horizon



Notes: This figure shows the evolution of net inflation spillovers across regions using the Diebold and Yilmaz (2012) methodology. Estimates are based on a fixed lag length 1 and a rolling window of 150 observations, while the forecast horizon H varies from 1 to 12. The solid black line represents the median net spillover across horizons, and the shaded area reflects the range between the minimum and maximum values.

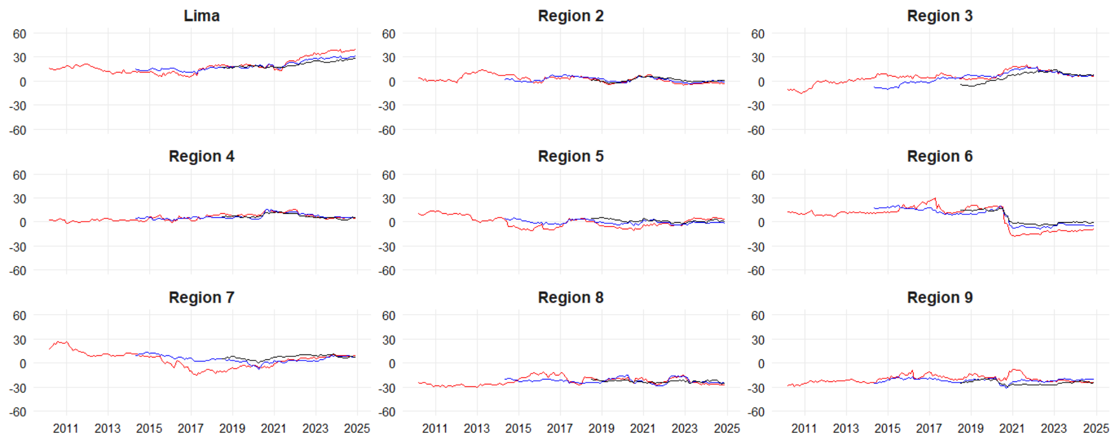
Figures 2 and 3 present additional robustness checks examining sensitivity to lag length specifications (L=1 to L=6) and rolling window sizes (100, 150, and 200 observations), respectively. Figure 2 illustrates the evolution of NET spillovers, with the solid black line denoting the median trajectory across lag specifications and the shaded area capturing the full range between minimum and maximum estimates. Across all variations, the fundamental pattern persists: Lima maintains its dominant net transmitter role, while Regions 8 and 9 consistently appear as net recipients. The stability of these relationships across diverse specifications reinforces confidence in our core empirical findings.

Figure 2: Robustness of Dynamic NET spillovers to Lag Length



Notes: This figure displays the evolution of net inflation spillovers across regions using the Diebold and Yilmaz (2012) methodology. Estimates are based on a fixed forecast horizon H = 3 and a rolling window of 150 observations, while the lag length varies from 1 to 6 to assess the robustness of results to alternative lag specifications.

Figure 3: Robustness of Dynamic NET spillovers to Rolling Window Size



Notes: This figure displays the evolution of net inflation spillovers across regions using the Diebold and Yilmaz (2012) methodology. Estimates are based on a fixed forecast horizon of 3 and a lag length of 1, while the rolling window size varies across 100, 150, and 200 observations.

City-Level Granular Analysis

As a comprehensive robustness check, we re-estimate spillover measures using monthly inflation data from Peru's 25 largest cities individually, rather than the nine regional aggregates employed in our main analysis. This granular approach tests whether our findings are sensitive to the spatial aggregation scheme and provides validation using the finest available geographic resolution.

Table 3 presents the city-level spillover results, revealing a total spillover index of 73.68%—virtually identical to our regional aggregate finding of 73.60[2]. This remarkable consistency demonstrates that our aggregation methodology preserves the essential spillover structure present in the underlying city-level data. Lima emerges as the dominant net transmitter at the city level with a NET spillover of 54.96 percentage points-even more pronounced than in regional analysis, confirming its central role in Peru's spatial inflation network

Cities corresponding to our peripheral regions (Iquitos, Puerto Maldonado) exhibit the largest negative NET spillovers (-40.00 and -36.18 percentage points), while cities in food-producing regions (Chiclayo, Huancayo) show significant positive transmission. These patterns align precisely with our regional findings, validating both our aggregation methodology and our economic interpretations of regional spillover roles.

[2]Estimation is based on a VAR model with one lag and a forecast horizon of three months.

Table 3: Time domain spillovers by cities

City	TO	FROM	NET
Abancay	71.8***	81.78***	-9.95
Arequipa	75.73***	79.17***	-3.44
Ayacucho	86.14***	82.63***	3.51
Cajamarca	56.93***	80.81***	-23.88***
Cerro de Pasco	100.32***	85.47***	14.85***
Chachapoyas	59.39***	83.13***	-23.75***
Chiclayo	102.58***	86.35***	16.22**
Chimbote	89.98***	85.17***	4.81
Cusco	72.52***	79.61***	-7.09
Huancavelica	78.48***	81.55***	-3.07
Huancayo	114.64***	83.15***	31.49***
Huánuco	82.07***	84.64***	-2.58
Huaraz	104.22***	85.70***	18.52**
Ica	87.45***	83.58***	3.87
Iquitos	33.65***	73.65***	-40.00***
Lima	141.91***	86.95***	54.96***
Moquegua	99.35***	83.00***	16.36*
Moyobamba	56.45***	82.66***	-26.21***
Piura	95.17***	85.16***	10.01
Pucallpa	62.95***	77.61***	-14.66
Puerto Maldonado	27.92***	64.10***	-36.18**
Puno	60.87***	76.65***	-15.78
Tacna	115.94***	81.57***	34.36***
Trujillo	94.55***	85.58***	8.97
Tumbes	70.88***	82.24***	-11.36
TSI		73.68***	

Notes: This table reports the time domain spillovers across cities of Peru. For the TSI, TO, FROM and NET spillover measures, we report statistical significance: *** significant at the 1% level, ** at the 5% level, and * at the 10% level.

C.2 Frequency Domain Approach

Alternative Frequency Band Specifications

To ensure our frequency-domain results are not dependent on the specific two-band decomposition employed in our main analysis, we implement a three-band specification following Istiak et al. (2021). This refined decomposition distinguishes between short-term $\left(\pi, \frac{\pi}{3}\right)$, medium-term $\left(\frac{\pi}{3}, \frac{\pi}{12}\right)$, and long-term $\left(\frac{\pi}{12}, 0\right)$ dynamics, corresponding approximately to 2-6 months, 6-24 months, and $24+$ months respectively.

Table 4 presents the three-band decomposition results, confirming and refining our main frequency-domain findings. In the short-term band, Lima appears as a statistically significant net recipient (NET = -3.24, $p<0.10$), consistent with our two-band analysis. However, the three-band decomposition reveals that Lima's transformation to dominance occurs primarily in the medium-term band (NET = 16.76, $p<0.01$), with continued strong transmission in the long-term band (NET = 10.48, $p<0.01$).

This three-band analysis provides crucial insights into the temporal structure of Lima's influence: its dominance emerges most strongly in the medium-term band (6-24 months), corresponding to monetary policy transmission cycles and business cycle frequencies. The persistence of significant transmission into the longest band (24+ months) confirms Lima's role in long-term inflation anchor formation and structural price determination.

Regions 8 and 9 maintain negative NET spillovers across all three bands, with particularly pronounced medium-term reception (-10.20 and -14.36 percentage points), suggesting that their structural dependence on external price determination operates most strongly at business cycle frequencies.

Table 4: Frequency domain spillovers in the short, medium and long run

	Lima	Region 2	Region 3	Region 4	Region 5	Region 6	Region 7	Region 8	Region 9	FROM
Short run: Bands (π - π/3)										
Lima	11,85	5,74	6,41	6,69	5,53	2,99	6,5	1,07	3,11	38.04***
Region 2	5,34	12,59	7,02	7,22	3,84	3,02	4,8	1,09	3,11	35.43***
Region 3	4,82	5,57	10,48	5,21	3,56	3,06	4,26	0,95	2,64	28.55***
Region 4	5,8	6,33	5,8	11,57	4,81	2,7	5,16	1,24	2,7	34.06***
Region 5	5,17	3,73	3,61	4,54	4,38	2,53	4,39	1,38	3,13	28.47***
Region 6	3,64	3,88	3,03	3,61	3,26	12,45	3,25	4,05	2,17	26.89***
Region 7	5,01	4,18	4,43	4,87	3,99	2,43	10,1	1,57	2,77	29.25***
Region 8	1,7	1,63	1,35	1,85	1,96	4,93	2,67	15,66	1,78	17.87***
Region 9	3,31	3,4	3,41	3,09	2,38	1,62	3,12	1,14	13,37	22.38***
TO	34.81***	34.47***	35.06***	37.07***	29.22***	22.26***	34.14***	12.50***	21.42***	
NET	-3.24*	-0.96	6.51***	3.01	0.75	-4.62	4.89*	-5.37**	-0.96	FSI = 28.99***
Medium run: Bands (π/3 - π/12)										
Lima	7,89	3,22	3,73	3,83	4,6	3,33	3,53	1,15	2,48	25.86***
Region 2	4,99	7,93	4,49	4,67	3,31	3,42	3,09	1,04	2,96	27.60***
Region 3	5,03	5,07	10,42	4,39	4,67	4,15	4,02	1,82	3,38	29.95***
Region 4	6,04	4,67	4,41	9,86	4,06	3,27	4,11	1,23	2,19	29.82***
Region 5	6,01	4,61	3,99	4,64	8,19	3,27	4,5	1,28	2,21	28.69***
Region 6	4,34	3,25	2,85	3,33	3,99	12,64	3,85	4,41	2,9	26.92***
Region 7	5,72	2,95	3,59	3,51	3,53	3,53	10,63	5,28	2,9	29.46***
Region 8	3,6	1,62	2,39	2,65	3,05	3,63	3,83	13,83	0,96	27.07***
Region 9	5,48	4,17	3,42	3,81	4,95	3,62	3,93	5,71	10,58	31.89***
TO	42.62***	28.16***	28.83***	30.69***	33.44***	31.70***	32.43***	16.87***	17.52***	
NET	16.76***	0.56	-3.55	0.72	-3.32	4.79	2.97	-10.20***	-14.36***	FSI = 29.13***
Long run: Bands (π/12 - 0)										
Lima	3,75	1,45	1,73	1,82	2,32	1,87	1,68	0,68	1,04	12.60***
Region 2	2,36	3,09	2,4	2,32	2,14	1,53	0,54	1,3	1,2	12.96***
Region 3	3,27	2,43	3,67	2,07	2,09	1,81	2,21	1,22	1,37	17.34***
Region 4	2,92	2,07	1,41	5,09	2,3	2,18	1,31	1,52	0,94	14.47***
Region 5	3,46	1,51	1,91	1,29	3,78	2,43	1,95	1,56	2,32	16.70***
Region 6	2,34	1,22	1,27	1,61	2,34	4,09	2,27	1,21	1,84	16.43***
Region 7	2,31	1,38	1,36	1,27	2,64	2,24	5,27	1,81	0,94	16.23***
Region 8	2,11	0,78	1,64	1,69	2,51	1,56	2,02	6,38	2,53	18.12***
Region 9	2,9	2,62	2,33	2,31	1,49	1,94	1,83	2,02	3,76	21.20***
TO	23.08***	13.09***	18.86***	15.96***	18.61***	18.65***	18.20***	10.38***	7.48***	
NET	10.48***	0.13	-2.47	1.49	-2.91	4.01	1.25	-6.87**	-9.93***	FSI = 15.57***

Notes: This table reports the spillovers in the frequency domain. For the FSI, TO, FROM and NET spillovers measures, we report its statistical significance: *** significant at the 1% level, ** at the 5% level, and * at the 10% level.

City-Level Frequency Analysis

We extend our frequency-domain robustness analysis to the city level, implementing the Baruník-Křehlík methodology on all 25 cities for both short-run and long-run bands. Tables 5 and 6 present selected results, confirming our regional frequency patterns at the finest spatial resolution[3].

[3] Estimation uses a VAR model with one lag and a forecast horizon of three months.

In the short-run band (Table 5), Lima shows a modest positive NET spillover (3.14) that is not statistically significant, while cities in agricultural regions (Chiclayo: 10.24, Huancayo: 3.30) exhibit stronger short-run transmission. This pattern aligns with our regional finding that food-producing areas drive immediate price pressures that subsequently affect Lima.

Table A5: Time domain spillovers by cities: short run $\left(\pi, \frac{\pi}{3}\right)$

City	TO	FROM	NET
Abancay	27.49***	27.63***	-0.14
Arequipa	26.62***	30.07***	-3.45
Ayacucho	31.27***	34.04***	-2.76
Cajamarca	29.23***	31.73***	-2.50
Cerro de Pasco	42.86***	27.73***	15.14***
Chachapoyas	26.76***	23.69***	3.07
Chiclayo	44.27***	34.03***	10.24***
Chimbote	44.04***	41.22***	2.83
Cusco	24.51***	26.47***	-1.96
Huancavelica	33.19***	39.83***	-6.63
Huancayo	36.07***	32.77***	3.30
Huánuco	44.29***	36.15***	8.14***
Huaraz	45.34***	35.08***	10.26***
Ica	29.78***	30.74***	-0.96
Iquitos	17.82***	24.05***	-6.23
Lima	51.12***	43.97***	3.14
Moquegua	35.53***	35.16***	0.37
Moyobamba	27.61***	28.64***	-1.03
Piura	45.90***	42.10***	3.81
Pucallpa	21.87***	32.78***	-10.92***
Puerto Maldonado	9.59***	22.13***	-12.54***
Puno	18.90***	22.96***	-4.06
Tacna	31.60***	43.14***	-11.54***
Trujillo	42.91***	39.71***	3.21
Tumbes	35.72***	38.51***	-2.79
TSI		32.97***	

Notes: This table reports the frequency domain spillovers across cities in the short run $\left(\pi, \frac{\pi}{3}\right)$. For the TSI, TO, FROM and NET spillover measures, we report statistical significance: *** significant at the 1% level, ** at the 5% level, and * at the 10% level.

In the long-run band (Table 6), Lima's dominance becomes overwhelming with a NET spillover of 47.87 percentage points ($p<0.01$)—substantially larger than any other city. Cities corresponding to peripheral regions (Iquitos: -34.20, Puerto Maldonado: -23.93) maintain their recipient roles, while secondary urban centers (Arequipa, Trujillo) show modest transmission capacity.

Table 6: Time domain spillovers by cities: long run $\left(\frac{\pi}{3},\ 0\right)$

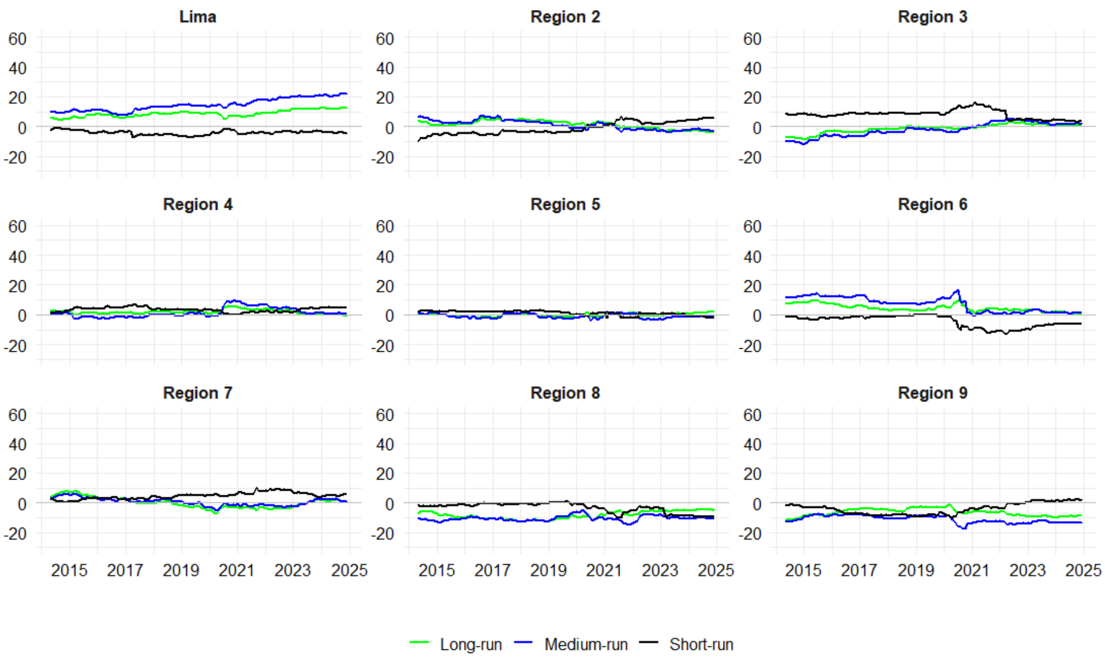
City	TO	FROM	NET
Abancay	44.88***	54.39***	-9.51
Arequipa	49.09***	49.19***	-0.10**
Ayacucho	55.52***	48.71***	6.80
Cajamarca	27.53***	49.19***	-21.65*
Cerro de Pasco	57.82***	57.92***	-0.09
Chachapoyas	32.71***	59.62***	-26.91***
Chiclayo	58.41***	52.38***	6.02**
Chimbote	45.77***	44.01***	1.76
Cusco	48.84***	53.36***	-4.51*
Huancavelica	45.59***	41.82***	3.78
Huancayo	79.44***	50.49***	28.95***
Huánuco	37.62***	48.62***	-11.01
Huaraz	59.39***	50.73***	8.65
Ica	58.19***	52.90***	5.29
Iquitos	15.69***	49.88***	-34.20***
Lima	90.87***	43.00***	47.87***
Moquegua	63.68***	47.91***	15.76
Moyobamba	28.64***	54.14***	-25.50***
Piura	48.54***	43.08***	5.46
Pucallpa	40.97***	44.97***	-4.00
Puerto Maldonado	18.66***	42.59***	-23.93**
Puno	42.71***	54.25***	-11.54
Tacna	84.87***	38.52***	46.35***
Trujillo	51.23***	45.92***	5.31
Tumbes	34.74***	43.79***	-9.05
TSI		48.86***	

Notes: This table reports the frequency domain spillovers across cities in the long run $\left(\frac{\pi}{3},\ 0\right)$. For the TSI, TO, FROM and NET spillover measures, we report statistical significance: *** significant at the 1% level, ** at the 5% level, and * at the 10% level.

Dynamic Frequency Robustness

Figure 4 presents the dynamic evolution of NET spillovers across our three-band frequency decomposition, confirming the temporal stability of frequency-dependent regional roles. Lima consistently displays negative NET spillovers in the short-term band throughout the sample period, while maintaining positive and generally increasing NET spillovers in both medium- and long-term bands. This temporal consistency across the entire sample period demonstrates that the frequency-dependent spillover patterns are structural features of Peru's regional economy rather than temporary phenomena.

Figure 4: Robustness of Dynamic NET spillovers in the short, medium and long run



Notes: This figure shows the short-term NET spillovers $\left(\pi, \frac{\pi}{3}\right)$, medium-term NET spillovers $\left(\frac{\pi}{3}, \frac{\pi}{12}\right)$ and long-term NET spillovers $\left(\frac{\pi}{12}, 0\right)$ estimated using the BK methodology. Results are based on a rolling window of 150 observations.